30-Minute Student Task

Python Student Attendance Management System — Solved

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1. Task Summary

The task is to build a simple Python-based Student Attendance Management System. The program accepts a student's name and attendance details, validates the entries, calculates the attendance percentage, and assigns an appropriate attendance status.

2. Requirement Checklist

Requirement	How it is satisfied
Accept student name	Uses input() to receive the student's name.
Accept attendance	Uses input() to receive attended and total classes.
Validate attendance	Checks that values are non-negative and attended classes do not exceed total classes.
Calculate percentage	Divides attended classes by total classes and multiplies by 100.
Assign status	Uses if / elif / else based on attendance percentage.
Display result	Prints total classes, attended classes, percentage, and status.

3. Complete Python Solution

```
name = input("Enter student name: ").strip()

while True:
    total_classes = int(input("Enter total number of classes: "))
    attended_classes = int(input("Enter number of classes attended: "))

    if total_classes > 0 and 0 <= attended_classes <= total_classes:
        break
    else:
        print("Enter valid attendance values.")

percentage = (attended_classes / total_classes) * 100

if percentage >= 90:
    status = "Excellent"
elif percentage >= 75:
    status = "Good"
elif percentage >= 60:
    status = "Average"
else:
    status = "Low"

print("\nStudent Attendance Result")
print("Name:", name)
print("Total Classes:", total_classes)
print("Classes Attended:", attended_classes)
print("Attendance Percentage:", round(percentage, 2))
print("Status:", status)
```

4. How the Program Works

INPUT	The program accepts the student's name, total number of classes, and attended classes.
VALIDATION	It checks that total classes are greater than zero and attended classes are within the valid range.
PROCESS	The attendance percentage is calculated using attended classes divided by total classes.
STATUS	An if/elif/else structure assigns an attendance status according to the percentage.
OUTPUT	The student's attendance details and final status are displayed clearly.

5. Test Cases and Expected Results

Test Input	Expected Result
Student: Rahul; Total: 100; Attended: 95	Attendance 95.0%; Status Excellent
Student: Priya; Total: 80; Attended: 64	Attendance 80.0%; Status Good
Student: Arun; Total: 50; Attended: 32	Attendance 64.0%; Status Average
Student: Ravi; Total: 60; Attended: 30	Attendance 50.0%; Status Low
Invalid: Total: 40; Attended: 45	Program asks for valid attendance values

6. Sample Demonstration

```
Enter student name: Rahul
Enter total number of classes: 100
Enter number of classes attended: 85

Student Attendance Result
Name: Rahul
Total Classes: 100
Classes Attended: 85
Attendance Percentage: 85.0
Status: Good
```

7. Advantages of the System

The system is simple, easy to understand, and suitable for beginners learning Python. It demonstrates input handling, loops, validation, arithmetic operations, and conditional statements. It also reduces manual calculation errors when determining attendance percentages and status.

8. Conclusion

The Python Student Attendance Management System demonstrates how basic Python concepts can be combined to solve a practical academic problem. The program accepts student information, validates attendance values, calculates attendance percentage, assigns a suitable status, and presents the result clearly. The system can be extended in the future to support multiple students, subject-wise attendance, monthly reports, low-attendance alerts, and file-based data storage.

9. Future Enhancements

- Store attendance records for multiple students.
- Maintain subject-wise attendance.
- Generate monthly or semester attendance reports.
- Add a warning when attendance falls below the required level.
- Save and retrieve records using a file or database.